

Power System Maintenance Checklists

15.2.1 Daily Checks Checklist

Facility: _____

Date: _____

Inspector: _____

Generator Fuel Level Verification

- ☐ [] Check fuel level gauge reading: _____% or _____gallons
- ☐ [] Verify fuel level is above 75% minimum threshold
- ☐ [] Inspect for fuel leaks around tanks, lines, and day tank
- ☐ [] Check fuel filter differential pressure indicators
- ☐ [] Verify automatic fuel transfer pump operation (if applicable)
- ☐ [] Record daily fuel consumption: _____gallons
- ☐ [] Notes: _____

UPS Status Indicator Check

- ☐ [] Verify all UPS systems display "Normal Operation" status
- ☐ [] Confirm input voltage is within spec: _____V (nominal: _____V)
- ☐ [] Confirm output voltage is within spec: _____V (nominal: _____V)
- ☐ [] Check battery charge status: _____%
- ☐ [] Verify no active alarms or warnings
- ☐ [] Check UPS load percentage: _____%
- ☐ [] Confirm cooling fans are operational
- ☐ [] Verify bypass status is normal
- ☐ [] Notes: _____

Alarm Panel Inspection

- ☐ [] Verify main alarm panel shows normal status
- ☐ [] Check for any active or latched alarms
- ☐ [] Test alarm panel indicator lights
- ☐ [] Verify communication with monitoring system
- ☐ [] Reset any acknowledged alarms
- ☐ [] Record any active alarms: _____
- ☐ [] Notes: _____

Battery Room Temperature Verification

- ☐ [] Record battery room temperature: _____ °F/°C
- ☐ [] Confirm temperature is between 68-77°F (20-25°C)
- ☐ [] Check HVAC system operation
- ☐ [] Verify exhaust fans are operational
- ☐ [] Check for any water leaks or condensation
- ☐ [] Ensure hydrogen detectors are operational (for flooded batteries)
- ☐ [] Notes: _____

Power Monitoring System Check

- ☐ [] Log into power monitoring system
- ☐ [] Verify all monitored devices are communicating
- ☐ [] Check for any threshold violations or alerts
- ☐ [] Review 24-hour power quality trends
- ☐ [] Confirm remote notification system is operational
- ☐ [] Verify data logging is functioning
- ☐ [] Notes: _____

Completed By: _____
Signature: _____ Time: _____

Supervisor Review: _____ Date: _____

15.2.2 Weekly Checks Checklist

Facility: _____

Week Starting: _____

Inspector: _____

Generator Test Run Verification

- ☐ [] Date of test run: _____
- ☐ [] Pre-check oil level and condition
- ☐ [] Pre-check coolant level and condition
- ☐ [] Verify fuel level before test: _____%
- ☐ [] Start generator and run for _____ minutes (minimum 15 minutes)
- ☐ [] Record voltage after stabilization: Phase A-B: _____V, B-C: _____V, C-A: _____V
- ☐ [] Record frequency after stabilization: _____Hz
- ☐ [] Record oil pressure during operation: _____psi
- ☐ [] Record coolant temperature during operation: _____°F/°C
- ☐ [] Record engine speed: _____RPM
- ☐ [] Check for unusual noise, vibration, or exhaust color
- ☐ [] Post-run inspection completed
- ☐ [] Verify automatic return to standby mode
- ☐ [] Notes: _____

Battery Terminal Inspection

- ☐ [] Inspect all visible battery terminals for corrosion
- ☐ [] Check for loose connections
- ☐ [] Look for bulging or damaged batteries
- ☐ [] Verify intercell connector tightness (spot check)
- ☐ [] Clean any corrosion using appropriate methods
- ☐ [] Check electrolyte levels (flooded batteries only)
- ☐ [] Measure ambient temperature in battery areas: _____°F/°C
- ☐ [] Notes: _____

Transfer Switch Status Verification

- ☐ [] Check each ATS for normal status indication
- ☐ [] Verify position indicators show utility power position
- ☐ [] Inspect for any signs of overheating or damage
- ☐ [] Check control power indicator
- ☐ [] Verify no alarms are present
- ☐ [] Confirm manual controls are accessible and operational

- ☐ [] Record any anomalies: _____
- ☐ [] Notes: _____

Power Quality Spot Checks

- ☐ [] Measure voltage at critical distribution points:
- ☐ Main distribution: Phase A-N: _____ V, B-N: _____ V, C-N: _____ V
 - ☐ Critical load panels: Phase A-N: _____ V, B-N: _____ V, C-N: _____ V
 - ☐ UPS output: Phase A-N: _____ V, B-N: _____ V, C-N: _____ V
- ☐ [] Check harmonic distortion at UPS input: THD = _____ %
- ☐ [] Check harmonic distortion at UPS output: THD = _____ %
- ☐ [] Measure frequency stability: _____ Hz
- ☐ [] Spot check ground-neutral voltage: _____ V
- ☐ [] Notes: _____

Cooling System Inspection

- ☐ [] Inspect all HVAC units serving electrical equipment
- ☐ [] Check cooling airflow at equipment intakes
- ☐ [] Verify temperature in UPS rooms: _____ °F/°C
- ☐ [] Verify temperature in server/mining equipment areas: _____ °F/°C
- ☐ [] Inspect air filters and note replacement needs
- ☐ [] Check for any water leaks from cooling systems
- ☐ [] Verify condensate drains are clear
- ☐ [] Notes: _____

Completed By: _____

Signature: _____ Date: _____

Supervisor Review: _____ Date: _____

15.2.3 Monthly Checks Checklist

Facility: _____

Month: _____

Inspector: _____

UPS Battery Inspection

- ☐ [] Date of inspection: _____
- ☐ [] Measure and record float voltage of each string: String 1: _____ V, String 2: _____ V
- ☐ [] Measure and record ripple voltage: _____ mV
- ☐ [] Perform battery voltage string comparison ($\pm 0.5V$ deviation max)
- ☐ [] Check individual battery voltage (spot check 10%): Within tolerance? Yes/No
- ☐ [] Inspect batteries for swelling or leakage
- ☐ [] Verify tight connections on all accessible terminals
- ☐ [] Check battery room ventilation operation
- ☐ [] Clean battery cabinets and surrounding area
- ☐ [] Record UPS battery temperature: _____ °F/°C
- ☐ [] Check battery monitoring system operation (if equipped)
- ☐ [] Notes: _____

Generator Load Test

- ☐ [] Date of test: _____
- ☐ [] Pre-check oil level and condition
- ☐ [] Pre-check coolant level and condition
- ☐ [] Inspect fuel system for leaks
- ☐ [] Check starting batteries' specific gravity or voltage: _____ V
- ☐ [] Start generator and run for _____ minutes (minimum 30 minutes)
- ☐ [] Transfer _____ % of building load to generator
- ☐ [] Record voltage under load: Phase A-B: _____ V, B-C: _____ V, C-A: _____ V
- ☐ [] Record frequency under load: _____ Hz
- ☐ [] Record load: Phase A: _____ A, Phase B: _____ A, Phase C: _____ A
- ☐ [] Record oil pressure during operation: _____ psi
- ☐ [] Record coolant temperature during operation: _____ °F/°C
- ☐ [] Check exhaust system for leaks
- ☐ [] Verify proper retransfer to utility and generator cooldown
- ☐ [] Notes: _____

Transfer Switch Operation Test

- ☐ [] Date of test: _____
- ☐ [] Notify affected departments before testing
- ☐ [] Pre-check switch condition and controls
- ☐ [] Initiate transfer to emergency source
- ☐ [] Record transfer time: _____seconds
- ☐ [] Verify proper operation of engine start signals
- ☐ [] Check transfer switch performance under load
- ☐ [] Measure voltage before, during, after transfer
- ☐ [] Initiate retransfer to normal source
- ☐ [] Record retransfer time: _____seconds
- ☐ [] Verify generator cooldown cycle
- ☐ [] Return all systems to normal operation
- ☐ [] Notes: _____

Power Distribution Inspection

- ☐ [] Check all distribution panels for heat, noise, or vibration
- ☐ [] Verify all circuit breakers are properly positioned
- ☐ [] Check panelboard schedules for accuracy
- ☐ [] Inspect for any tripped breakers or blown fuses
- ☐ [] Verify proper panel labeling
- ☐ [] Check for any signs of moisture or corrosion
- ☐ [] Manually verify operation of critical breakers (if scheduled)
- ☐ [] Record load balance: Phase A: _____%, Phase B: _____%, Phase C: _____%
- ☐ [] Notes: _____

Thermal Scanning of Critical Connections

- ☐ [] Date of scan: _____
- ☐ [] Equipment used: _____
- ☐ [] Scan main distribution connections
- ☐ [] Scan UPS input and output connections
- ☐ [] Scan generator output connections
- ☐ [] Scan critical transfer switches
- ☐ [] Scan primary mining equipment power connections
- ☐ [] Record maximum temperature detected: _____°F/°C at _____
- ☐ [] Record ambient temperature during scan: _____°F/°C

☐ [] Document any connections exceeding 15°F/°C above ambient

☐ [] Corrective actions required? Yes/No (if yes, detail below)

☐ [] Notes: _____

Completed By: _____

Signature: _____ Date: _____

Supervisor Review: _____ Date: _____

15.2.4 Quarterly Checks Checklist

Facility: _____

Quarter: _____

Inspector: _____

Comprehensive UPS Testing

- ☐ [] Date of testing: _____
- ☐ [] Review alarm logs and event history
- ☐ [] Verify proper operation in all modes:
 - ☐ [] Normal operation
 - ☐ [] Battery operation
 - ☐ [] Bypass operation
 - ☐ [] Return to normal
- ☐ [] Perform manual transfer to bypass and back
- ☐ [] Check all indicator lights and display
- ☐ [] Verify remote monitoring connectivity
- ☐ [] Measure input parameters:
 - ☐ Voltage: Phase A-N: _____ V, B-N: _____ V, C-N: _____ V
 - ☐ Current: Phase A: _____ A, Phase B: _____ A, Phase C: _____ A
 - ☐ Frequency: _____ Hz
 - ☐ Power factor: _____
- ☐ [] Measure output parameters:
 - ☐ Voltage: Phase A-N: _____ V, B-N: _____ V, C-N: _____ V
 - ☐ Current: Phase A: _____ A, Phase B: _____ A, Phase C: _____ A
 - ☐ Frequency: _____ Hz
 - ☐ Load percentage: _____ %
- ☐ [] Perform short-duration battery test (5 minutes)
- ☐ [] Check ventilation and cooling systems
- ☐ [] Inspect capacitors for bulging or leakage
- ☐ [] Clean interior and exterior of cabinets
- ☐ [] Review and update firmware (if required)
- ☐ [] Notes: _____

Generator Load Bank Testing

- ☐ [] Date of testing: _____
- ☐ [] Generator manufacturer/model: _____
- ☐ [] Load bank capacity: _____ kW
- ☐ [] Pre-inspection completed:
 - ☐ [] Oil level and condition

- ☐ ☐ Coolant level and condition
- ☐ ☐ Fuel level: _____%
- ☐ ☐ Air filters
- ☐ ☐ Starting batteries condition
- ☐ ☐ Exhaust system integrity
- ☐ ☐ Start generator and warm up for _____ minutes
- ☐ ☐ Apply load in steps:
 - ☐ 25% load: _____kW for _____ minutes
 - ☐ 50% load: _____kW for _____ minutes
 - ☐ 75% load: _____kW for _____ minutes
 - ☐ 100% load: _____kW for _____ minutes
- ☐ ☐ Record readings at 100% load:
 - ☐ Voltage: Phase A-B: _____V, B-C: _____V, C-A: _____V
 - ☐ Current: Phase A: _____A, Phase B: _____A, Phase C: _____A
 - ☐ Frequency: _____Hz
 - ☐ Oil pressure: _____psi
 - ☐ Coolant temperature: _____°F/°C
 - ☐ Exhaust temperature: _____°F/°C
 - ☐ Ambient temperature: _____°F/°C
- ☐ ☐ Record fuel consumption rate: _____gallons/hour
- ☐ ☐ Check for unusual vibration, noise, or exhaust color
- ☐ ☐ Perform proper cooldown procedure
- ☐ ☐ Post-inspection completed
- ☐ ☐ Notes: _____

Protection System Verification

- ☐ ☐ Date of verification: _____
- ☐ ☐ Test GFCI outlets in critical areas
- ☐ ☐ Check surge suppression devices status indicators
- ☐ ☐ Verify proper operation of emergency stops
- ☐ ☐ Test ground fault monitoring systems
- ☐ ☐ Verify settings on electronic trip circuit breakers
- ☐ ☐ Check calibration of protective relays (if applicable)
- ☐ ☐ Test operation of shunt trip mechanisms (if scheduled)
- ☐ ☐ Verify coordination with upstream/downstream devices
- ☐ ☐ Notes: _____

Grounding System Inspection

- ☐ [] Date of inspection: _____
- ☐ [] Visually inspect all accessible ground connections
- ☐ [] Measure ground resistance: _____ ohms (target: <5 ohms)
- ☐ [] Check ground bus continuity in all equipment
- ☐ [] Verify equipment grounding conductor connections
- ☐ [] Measure neutral-to-ground voltage: _____ V
- ☐ [] Inspect lightning protection system components
- ☐ [] Check for signs of ground conductor damage
- ☐ [] Notes: _____

Complete Power Quality Analysis

- ☐ [] Date of analysis: _____
- ☐ [] Equipment used: _____
- ☐ [] Duration of monitoring: _____ hours
- ☐ [] Measurement points:
 - ☐ Main service entrance
 - ☐ UPS input
 - ☐ UPS output
 - ☐ Critical mining equipment panels
- ☐ [] Parameters measured:
 - ☐ Voltage variations (min/max): _____ V / _____ V
 - ☐ Frequency variations (min/max): _____ Hz / _____ Hz
 - ☐ Voltage harmonic distortion (THD): _____ %
 - ☐ Current harmonic distortion (THD): _____ %
 - ☐ Voltage sags/swells: _____ events
 - ☐ Transients: _____ events
 - ☐ Power factor: _____
- ☐ [] Data recorded and archived
- ☐ [] Analysis performed and documented
- ☐ [] Corrective actions identified? Yes/No (if yes, detail below)
- ☐ [] Notes: _____

Completed By: _____

Signature: _____ Date: _____

Supervisor Review: _____ Date: _____

15.2.5 Annual Checks Checklist

Facility: _____

Year: _____

Inspector: _____

UPS Battery Capacity Testing

- ☐ [] Date of testing: _____
- ☐ [] Testing method: ☐ Discharge test ☐ Impedance test ☐ Other: _____
- ☐ [] Battery manufacturer: _____ Model: _____
- ☐ [] Installation date: _____ Rated capacity: _____ Ah
- ☐ [] For discharge test:
 - ☐ [] Pre-test voltage readings taken for all batteries
 - ☐ [] Load bank connected: _____ kW
 - ☐ [] Discharge rate: _____ A
 - ☐ [] Target runtime: _____ minutes
 - ☐ [] Actual runtime achieved: _____ minutes
 - ☐ [] Lowest cell voltage at end of test: _____ V
 - ☐ [] Temperature during test: _____ °F/°C
- ☐ [] For impedance test:
 - ☐ [] Baseline values available for comparison? Yes/No
 - ☐ [] Number of batteries exceeding 20% increase from baseline: _____
 - ☐ [] Number of batteries exceeding 50% increase from baseline: _____
- ☐ [] Post-test analysis completed
- ☐ [] Recharge profile verified
- ☐ [] Replacement recommendations: _____
- ☐ [] Notes: _____

Generator Full-Load Endurance Testing

- ☐ [] Date of testing: _____
- ☐ [] Generator manufacturer/model: _____ Capacity: _____ kW
- ☐ [] Test method: ☐ Building load ☐ Load bank ☐ Combination
- ☐ [] Pre-inspection completed
- ☐ [] Full load applied: kW (% of rated capacity)
- ☐ [] Duration of full-load test: _____ hours (minimum 4 hours)
- ☐ [] Readings recorded at 30-minute intervals (attach separate log)
- ☐ [] Record stabilized parameters:
 - ☐ Voltage: Phase A-B: _____ V, B-C: _____ V, C-A: _____ V
 - ☐ Current: Phase A: _____ A, Phase B: _____ A, Phase C: _____ A

- ☐ Frequency: _____ Hz
- ☐ Power factor: _____
- ☐ Oil pressure: _____ psi
- ☐ Oil temperature: _____ °F/°C
- ☐ Coolant temperature: _____ °F/°C
- ☐ Exhaust temperature: _____ °F/°C
- ☐ Ambient temperature: _____ °F/°C
- ☐ [] Fuel consumption rate: _____ gallons/hour
- ☐ [] Oil consumption rate: _____ quarts/hour
- ☐ [] Governor response test performed
- ☐ [] Load rejection test performed
- ☐ [] Proper cooldown procedure followed
- ☐ [] Post-test inspection completed
- ☐ [] Notes: _____

Infrared Scanning of All Electrical Equipment

- ☐ [] Date of scanning: _____
- ☐ [] Equipment used: _____ Calibration date: _____
- ☐ [] Qualified thermographer: _____
- ☐ [] Systems scanned:
 - ☐ [] Main switchgear
 - ☐ [] Distribution panels
 - ☐ [] UPS systems
 - ☐ [] Battery connections
 - ☐ [] Transfer switches
 - ☐ [] Generator connections
 - ☐ [] Motor control centers
 - ☐ [] Transformers
 - ☐ [] Bus ducts
 - ☐ [] Critical mining equipment connections
- ☐ [] Number of critical findings (>50°F/28°C above ambient): _____
- ☐ [] Number of serious findings (30-50°F/17-28°C above ambient): _____
- ☐ [] Number of intermediate findings (15-30°F/8-17°C above ambient): _____
- ☐ [] Number of minor findings (<15°F/8°C above ambient): _____
- ☐ [] Corrective actions documented and prioritized
- ☐ [] Complete report filed separately
- ☐ [] Notes: _____

Complete System Performance Evaluation

- ☐ [] Date of evaluation: _____
- ☐ [] Review annual power availability: _____%
- ☐ [] Review number of unplanned outages: _____
- ☐ [] Review average outage duration: _____minutes
- ☐ [] Review total downtime: _____hours
- ☐ [] Power quality compliance percentage: _____%
- ☐ [] UPS efficiency: _____%
- ☐ [] Generator reliability (successful starts): _____%
- ☐ [] Transfer system reliability: _____%
- ☐ [] System capacity utilization: _____%
- ☐ [] Energy consumption analysis completed
- ☐ [] Maintenance compliance analysis completed
- ☐ [] Equipment lifecycle status reviewed
- ☐ [] System expansion requirements assessed
- ☐ [] Technology upgrade opportunities identified
- ☐ [] Budget forecast for next year completed
- ☐ [] Performance improvement recommendations documented
- ☐ [] Notes: _____

Protection Coordination Verification

- ☐ [] Date of verification: _____
- ☐ [] Protective device settings reviewed
- ☐ [] Time-current curves analyzed for proper coordination
- ☐ [] Changes in system configuration identified since last review
- ☐ [] Changes in load characteristics identified since last review
- ☐ [] Arc flash analysis reviewed and updated
- ☐ [] Arc flash labels verified for accuracy
- ☐ [] Selective coordination verified for emergency systems
- ☐ [] Protective relay testing completed (if applicable)
- ☐ [] Trip unit testing completed for critical breakers
- ☐ [] Settings adjusted as needed? Yes/No (if yes, detail below)
- ☐ [] Documentation updated with new settings
- ☐ [] Notes: _____

Completed By: _____
Signature: _____ Date: _____

Supervisor Review: _____ Date: _____
Engineering Review: _____ Date: _____

15.3.1 Power System Inspection Forms

Daily System Inspection Form

BITCOIN MINING FACILITY POWER SYSTEMS DAILY INSPECTION FORM

Facility Name: _____

Date: _____

Inspector Name: _____

Shift: ☐ Morning ☐ Afternoon ☐ Night

1. GENERATOR SYSTEM STATUS

Item	Status	Comments
Fuel Level (%)	_____ %	
Fuel System Alarms	☐ OK ☐ Alarm	
Engine Coolant Level	☐ OK ☐ Low	
Oil Level	☐ OK ☐ Low	
Battery Charger Status	☐ OK ☐ Fault	
Control Panel Status	☐ OK ☐ Alarm	
Abnormal Noises/Vibration	☐ None ☐ Present	
Leaks (Oil, Fuel, Coolant)	☐ None ☐ Present	

2. UPS SYSTEM STATUS

Item	Status	Comments
UPS Operation Mode	☐ Normal ☐ Bypass ☐ Battery	
Input Voltage L1-N: ____ V L2-N: ____ V L3-N: ____ V	☐ In Range ☐ Out of Range	
Output Voltage L1-N: ____ V L2-N: ____ V L3-N: ____ V	☐ In Range ☐ Out of Range	
Load Level (%)	_____ %	

Battery Charge Level (%)	_____ %	
Active Alarms	☐ None ☐ Present	
Abnormal Noise/Heat	☐ None ☐ Present	
Bypass Status	☐ Available ☐ Unavailable	

3. BATTERY ROOM CONDITIONS

Item	Status	Comments
Room Temperature	_____ °F/°C	
Ventilation System	☐ OK ☐ Fault	
Hydrogen Monitoring	☐ OK ☐ Alarm	
Water/Moisture	☐ None ☐ Present	
Security/Access Control	☐ Secure ☐ Issue	

4. POWER DISTRIBUTION

Item	Status	Comments
Main Distribution Panel Status	☐ Normal ☐ Issue	
Critical Load Panel Status	☐ Normal ☐ Issue	
Circuit Breaker Positions	☐ Normal ☐ Tripped	
Abnormal Heat/Noise	☐ None ☐ Present	

5. MONITORING SYSTEMS

Item	Status	Comments
BMS/Power Monitoring Status	☐ Operational ☐ Fault	
Active System Alarms	☐ None ☐ Present: _____	
Remote Monitoring Connection	☐ Connected ☐ Disconnected	

6. ENVIRONMENTAL SYSTEMS FOR POWER EQUIPMENT

Item	Status	Comments
HVAC Systems	☐ Operational ☐ Fault	
Temperature in Power Rooms	_____ °F/°C	
Air Filtration	☐ Clean ☐ Needs Service	
Water Detection Systems	☐ OK ☐ Alarm	

7. INCIDENTS AND OBSERVATIONS

8. ACTIONS TAKEN

9. RECOMMENDED FOLLOW-UP

Inspector Signature: _____ Time Completed: _____

Supervisor Review: _____ Date: _____

Weekly Equipment Check Form

BITCOIN MINING FACILITY POWER SYSTEMS WEEKLY EQUIPMENT CHECK FORM

Facility Name: _____

Week of: _____

Inspector Name: _____

Date Performed: _____

1. GENERATOR WEEKLY CHECKS

Check Item	Result	Acceptance Criteria	Pass/Fail	Comments
No-Load Test Run	☐ Completed ☐ Not Completed			
Run Duration	_____ minutes	Minimum 15 minutes	☐ Pass ☐ Fail	
Voltage Output	L1-L2: _____ V L2-L3: _____ V L3-L1: _____ V	Within $\pm 5\%$ of nominal	☐ Pass ☐ Fail	
Frequency	_____ Hz	60 ± 0.5 Hz or 50 ± 0.5 Hz	☐ Pass ☐ Fail	
Oil Pressure	_____ psi	Within manufacturer specs	☐ Pass ☐ Fail	
Coolant Temperature	_____ °F/°C	Within manufacturer specs	☐ Pass ☐ Fail	
Battery Voltage	_____ V	≥ 12.5 V (12V system) or ≥ 25 V (24V system)	☐ Pass ☐ Fail	
Fuel Lines/Connections		No leaks	☐ Pass ☐ Fail	
Air Filter Condition		Clean, not restricted	☐ Pass ☐ Fail	
Exhaust System		No leaks, secure	☐ Pass ☐ Fail	
Air Intake System		Clear, no restrictions	☐ Pass ☐ Fail	

Belts and Hoses		Good condition, proper tension	☐ Pass ☐ Fail	
Vibration		Normal, no excessive vibration	☐ Pass ☐ Fail	
Control Panel Operation		All functions normal	☐ Pass ☐ Fail	

2. UPS WEEKLY CHECKS

Check Item	Result	Acceptance Criteria	Pass/Fail	Comments
Visual Inspection		No physical damage	☐ Pass ☐ Fail	
Cooling Fans		All operational	☐ Pass ☐ Fail	
Air Intake/Exhaust		Clear, unobstructed	☐ Pass ☐ Fail	
Dust/Debris		Clean	☐ Pass ☐ Fail	
Event Log Review		No critical events	☐ Pass ☐ Fail	
Alarm Status		No active alarms	☐ Pass ☐ Fail	
Front Panel Operation		All indicators functional	☐ Pass ☐ Fail	
Unusual Noise		None	☐ Pass ☐ Fail	
Communication Ports		Properly connected	☐ Pass ☐ Fail	

3. BATTERY WEEKLY CHECKS

Check Item	Result	Acceptance Criteria	Pass/Fail	Comments
Terminal Connections		Tight, no corrosion	☐ Pass ☐ Fail	
Battery Case Condition		No cracks, bulging, leakage	☐ Pass ☐ Fail	
Electrolyte Level (if applicable)		Within range	☐ Pass ☐ Fail	

Terminal Temperature (spot check)		No hot terminals	☐ Pass ☐ Fail	
Battery Room Cleanliness		Clean, no debris	☐ Pass ☐ Fail	
Ventilation System		Operational	☐ Pass ☐ Fail	

4. TRANSFER SWITCH WEEKLY CHECKS

Check Item	Result	Acceptance Criteria	Pass/Fail	Comments
Visual Inspection		No physical damage	☐ Pass ☐ Fail	
Position Indicators		Correct indication	☐ Pass ☐ Fail	
Control Power		Present	☐ Pass ☐ Fail	
Abnormal Noise/Heat		None	☐ Pass ☐ Fail	
Dust/Debris		Clean	☐ Pass ☐ Fail	
External Connections		Secure	☐ Pass ☐ Fail	

5. POWER QUALITY SPOT CHECKS

Location	Voltage	Current	Power Factor	Harmonics (THD)	Comments
Main Input	L1-N: ____V L2-N: ____V L3-N: ____V	L1: ____A L2: ____A L3: ____A	_____	_____%	
UPS Output	L1-N: ____V L2-N: ____V L3-N: ____V	L1: ____A L2: ____A L3: ____A	_____	_____%	
Critical Panel	L1-N: ____V L2-N: ____V L3-N: ____V	L1: ____A L2: ____A L3: ____A	_____	_____%	

6. NOTABLE FINDINGS AND OBSERVATIONS

7. MAINTENANCE ACTIONS TAKEN

8. REQUIRED FOLLOW-UP ACTIONS

Action Required	Priority (High/Med/Low)	Assigned To	Target Date	Comments

Inspector Signature: _____

Date Completed: _____

Supervisor Review: _____

Date: _____

Monthly Maintenance Verification Form

BITCOIN MINING FACILITY POWER SYSTEMS MONTHLY MAINTENANCE VERIFICATION FORM

Facility Name: _____

Month/Year: _____

Inspector Name: _____

Date Performed: _____

1. GENERATOR MONTHLY MAINTENANCE

Maintenance Task	Completed	Findings/Measurements	Comments
Load Test	☐ Yes ☐ No		
- Duration		_____ minutes	Minimum 30 minutes
- Load Percentage		_____ %	
- Voltage Under Load		L1-L2: _____ V L2-L3: _____ V L3-L1: _____ V	
- Frequency Under Load		_____ Hz	
- Oil Pressure		_____ psi	
- Coolant Temperature		_____ °F/°C	
Fuel System	☐ Yes ☐ No		
- Fuel Quality Check			
- Water Separator Check			
- Fuel Filter Inspection			
Cooling System	☐ Yes ☐ No		
- Coolant Level/Quality			

- Radiator Inspection			
- Fan Belt Tension			
Starting System	☐ Yes ☐ No		
- Battery Condition			
- Battery Charger Output		_____ V, _____ A	
- Starter Inspection			
Lubrication System	☐ Yes ☐ No		
- Oil Level/Condition			
- Oil Filter Inspection			
- Oil Leaks Check			
Air Intake System	☐ Yes ☐ No		
- Air Filter Inspection			
- Intake Piping Check			
Exhaust System	☐ Yes ☐ No		
- Exhaust Leaks Check			
- Exhaust Restriction			
Control System	☐ Yes ☐ No		
- Alarm Functions Test			
- Safety Shutdown Test			

- Transfer Operation			
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2. UPS MONTHLY MAINTENANCE

Maintenance Task	Completed	Findings/Measurements	Comments
Visual Inspection	☐ Yes ☐ No		
- External Cabinet			
- Ventilation			
- Display/Controls			
Operational Check	☐ Yes ☐ No		
- Input Voltage		L1-N: ____ V L2-N: ____ V L3-N: ____ V	
- Output Voltage		L1-N: ____ V L2-N: ____ V L3-N: ____ V	
- Input/Output Frequency		In: ____ Hz Out: ____ Hz	
- Load Percentage		____ %	
Battery System	☐ Yes ☐ No		
- Battery Voltage		____ V	
- Charge Current		____ A	
- Battery Temperature		____ °F/°C	
- Terminal Inspection			
Event Log Review	☐ Yes ☐ No		
- Alarm History			
- Performance Trends			
Cooling System	☐ Yes ☐ No		
- Fan Operation			

- Air Filter Condition			
System Test	☐ Yes ☐ No		
- Transfer to Battery			
- Return to Normal			
- Bypass Operation			

3. TRANSFER SWITCH MONTHLY MAINTENANCE

Maintenance Task	Completed	Findings/Measurements	Comments
Visual Inspection	☐ Yes ☐ No		
- Enclosure Integrity			
- Internal Components			
- Power Connections			
Operational Test	☐ Yes ☐ No		
- Transfer Time		_____ seconds	
- Return Time		_____ seconds	
- Engine Start Signal			
- Position Indicators			
Control System	☐ Yes ☐ No		
- Logic Control Test			
- Timing Functions			
- Manual Controls			
Electrical Connections	☐ Yes ☐ No		
- Terminal Tightness			
- Signs of Overheating			
- Insulation Condition			

Mechanical Components	☐ Yes ☐ No		
- Moving Parts Lubrication			
- Mechanism Operation			
- Interlocks Function			

4. POWER DISTRIBUTION INSPECTION

Maintenance Task	Completed	Findings/Measurements	Comments
Main Distribution	☐ Yes ☐ No		
- Visual Inspection			
- Load Balance		L1: _____% L2: _____% L3: _____%	
- Thermal Scan			
PDU's/RPPs	☐ Yes ☐ No		
- Visual Inspection			
- Load Readings			
- Branch Circuit Check			
Circuit Breakers	☐ Yes ☐ No		
- Visual Inspection			
- Manual Operation (if scheduled)			
- Connection Tightness			
Grounding System	☐ Yes ☐ No		
- Connection Integrity			
- Corrosion Check			
- Ground Bus Inspection			

5. THERMAL SCANNING RESULTS

Equipment	Maximum Temperature	Ambient Temperature	Temperature Rise	Action Required
	____ °F/°C	____ °F/°C	____ °F/°C	☐ None ☐ Monitor ☐ Repair
	____ °F/°C	____ °F/°C	____ °F/°C	☐ None ☐ Monitor ☐ Repair
	____ °F/°C	____ °F/°C	____ °F/°C	☐ None ☐ Monitor ☐ Repair
	____ °F/°C	____ °F/°C	____ °F/°C	☐ None ☐ Monitor ☐ Repair
	____ °F/°C	____ °F/°C	____ °F/°C	☐ None ☐ Monitor ☐ Repair

6. SIGNIFICANT FINDINGS AND OBSERVATIONS

7. MAINTENANCE ACTIONS COMPLETED

8. REQUIRED FOLLOW-UP ACTIONS

Action Required	Priority (High/Med/Low)	Assigned To	Target Date	Comments

9. PARTS/MATERIALS USED

Part Description	Part Number	Quantity	Equipment/Location

Inspector Signature: _____

Date Completed: _____

Supervisor Review: _____

Date: _____

Quarterly Comprehensive Inspection Form

BITCOIN MINING FACILITY POWER SYSTEMS QUARTERLY COMPREHENSIVE INSPECTION FORM

Facility Name: _____

Quarter/Year: _____

Inspector Name: _____

Date Performed: _____

1. UPS COMPREHENSIVE TESTING

Test/Inspection Item	Results	Pass/Fail	Comments
Full Functional Testing			
- Normal Mode Operation		☐ Pass ☐ Fail	
- Battery Mode Operation		☐ Pass ☐ Fail	
- Bypass Mode Operation		☐ Pass ☐ Fail	
- Return to Normal Operation		☐ Pass ☐ Fail	
Performance Measurements			
- Input Voltage Quality		☐ Pass ☐ Fail	
- Output Voltage Regulation		☐ Pass ☐ Fail	
- Frequency Regulation		☐ Pass ☐ Fail	
- Harmonic Distortion		☐ Pass ☐ Fail	
- Efficiency		☐ Pass ☐ Fail	
Battery System Testing			
- Runtime Test (_____ minutes)		☐ Pass ☐ Fail	
- Individual Battery Inspection		☐ Pass ☐ Fail	
- Connection Integrity		☐ Pass ☐ Fail	
- Charger Performance		☐ Pass ☐ Fail	
Cooling System			

- Fan Operation		☐ Pass ☐ Fail	
- Temperature Measurements		☐ Pass ☐ Fail	
- Air Flow Verification		☐ Pass ☐ Fail	
Control System			
- Display Functions		☐ Pass ☐ Fail	
- Alarm Functions		☐ Pass ☐ Fail	
- Communication Interfaces		☐ Pass ☐ Fail	
- Firmware Version		☐ Pass ☐ Fail	
Internal Components			
- Capacitor Inspection		☐ Pass ☐ Fail	
- Power Module Inspection		☐ Pass ☐ Fail	
- Internal Connections		☐ Pass ☐ Fail	
- Filter Condition		☐ Pass ☐ Fail	

2. GENERATOR LOAD BANK TESTING

Test/Inspection Item	Results	Pass/Fail	Comments
Load Bank Test			
- 25% Load Test (_____ minutes)		☐ Pass ☐ Fail	
- 50% Load Test (_____ minutes)		☐ Pass ☐ Fail	
- 75% Load Test (_____ minutes)		☐ Pass ☐ Fail	
- 100% Load Test (_____ minutes)		☐ Pass ☐ Fail	
Performance Under Load			
- Voltage Stability		☐ Pass ☐ Fail	
- Frequency Stability		☐ Pass ☐ Fail	
- Oil Pressure		☐ Pass ☐ Fail	

- Coolant Temperature		☐ Pass ☐ Fail	
- Exhaust Temperature		☐ Pass ☐ Fail	
- Engine Performance		☐ Pass ☐ Fail	
Cooling System Under Load			
- Radiator Performance		☐ Pass ☐ Fail	
- Fan Operation		☐ Pass ☐ Fail	
- Temperature Stabilization		☐ Pass ☐ Fail	
Fuel System Under Load			
- Fuel Consumption Rate		☐ Pass ☐ Fail	
- Fuel Delivery		☐ Pass ☐ Fail	
- Return Line Operation		☐ Pass ☐ Fail	
Control System			
- Response to Load Changes		☐ Pass ☐ Fail	
- Protection Functions		☐ Pass ☐ Fail	
- Monitoring Accuracy		☐ Pass ☐ Fail	
Exhaust System			
- Back Pressure		☐ Pass ☐ Fail	
- Emissions (if tested)		☐ Pass ☐ Fail	
- Leakage		☐ Pass ☐ Fail	
Recovery After Test			
- Cooldown Performance		☐ Pass ☐ Fail	
- Return to Standby		☐ Pass ☐ Fail	

3. PROTECTION SYSTEM VERIFICATION

System/Component	Test Performed	Results	Pass/Fail	Comments
Ground Fault Protection			☐ Pass ☐ Fail	
Circuit Breaker Testing			☐ Pass ☐ Fail	
Surge Protection Devices			☐ Pass ☐ Fail	
Lightning Protection System			☐ Pass ☐ Fail	
Emergency Power Off System			☐ Pass ☐ Fail	
Overcurrent Protection			☐ Pass ☐ Fail	
Undervoltage Protection			☐ Pass ☐ Fail	
Overvoltage Protection			☐ Pass ☐ Fail	
Protective Relays			☐ Pass ☐ Fail	
Coordination Check			☐ Pass ☐ Fail	

4. GROUNDING SYSTEM INSPECTION

Test/Inspection Item	Results	Pass/Fail	Comments
Earth Resistance Test	_____ Ohms	☐ Pass ☐ Fail	Target: <5 Ohms
Equipment Grounding		☐ Pass ☐ Fail	
Signal Reference Grid		☐ Pass ☐ Fail	
Lightning Protection		☐ Pass ☐ Fail	
Ground Bus Connections		☐ Pass ☐ Fail	
Bonding Connections		☐ Pass ☐ Fail	

Isolated Grounds		☐ Pass ☐ Fail	
Corrosion Inspection		☐ Pass ☐ Fail	
Ground Cable Integrity		☐ Pass ☐ Fail	

5. POWER QUALITY ANALYSIS

Parameter	Measurement Location	Results	Acceptable Range	Pass/Fail	Comments
Voltage Variation				☐ Pass ☐ Fail	
Frequency Variation				☐ Pass ☐ Fail	
Voltage Harmonic Distortion				☐ Pass ☐ Fail	
Current Harmonic Distortion				☐ Pass ☐ Fail	
Voltage Sags/Swells				☐ Pass ☐ Fail	
Transients				☐ Pass ☐ Fail	
Power Factor				☐ Pass ☐ Fail	
Phase Balance				☐ Pass ☐ Fail	
Neutral Current				☐ Pass ☐ Fail	

6. EQUIPMENT CONDITION ASSESSMENT

System	Current Condition	Remaining Life Estimate	Recommended Actions
UPS System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Generator System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Battery System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Transfer Switches	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Distribution System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Cooling System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Monitoring System	☐ Excellent ☐ Good ☐ Fair ☐ Poor		

7. SYSTEM PERFORMANCE METRICS (QUARTERLY)

Metric	Current Quarter	Previous Quarter	Target	Trend
System Availability (%)				☐ Improved ☐ Steady ☐ Declined
Number of Outages				☐ Improved ☐ Steady ☐ Declined
Average Outage Duration				☐ Improved ☐ Steady ☐ Declined
UPS Efficiency (%)				☐ Improved ☐ Steady ☐ Declined
Generator Reliability (%)				☐ Improved ☐ Steady ☐ Declined

Power Quality Compliance (%)				☐ Improved ☐ Steady ☐ Declined
Energy Usage (kWh)				☐ Improved ☐ Steady ☐ Declined

8. MAJOR FINDINGS AND OBSERVATIONS

9. COMPREHENSIVE RECOMMENDATIONS

10. PRIORITY ACTION ITEMS

Action Item	Priority (High/Med/Low)	Responsible Party	Target Completion	Budget Estimate

Primary Inspector: _____

Date: _____

Technical Review: _____

Date: _____

Management Review: _____

Date: _____

Annual System Audit Form

BITCOIN MINING FACILITY POWER SYSTEMS ANNUAL SYSTEM AUDIT FORM

Facility Name: _____

Year: _____

Lead Auditor: _____

Audit Date(s): _____

Audit Team Members: _____

1. SYSTEM OVERVIEW

System Description	Manufacturer	Model	Installation Date	Last Major Service	Design Life	Current Age
UPS System						
Generator System						
Battery System						
Transfer Switches						
Distribution System						
Cooling System						
Monitoring System						

2. DOCUMENTATION REVIEW

Document Type	Available	Current	Comments
Single-Line Diagrams	☐ Yes ☐ No	☐ Yes ☐ No	
Equipment Specifications	☐ Yes ☐ No	☐ Yes ☐ No	
Operation Manuals	☐ Yes ☐ No	☐ Yes ☐ No	
Maintenance Records	☐ Yes ☐ No	☐ Yes ☐ No	

Test Reports	☐ Yes ☐ No	☐ Yes ☐ No	
Incident Reports	☐ Yes ☐ No	☐ Yes ☐ No	
Training Records	☐ Yes ☐ No	☐ Yes ☐ No	
Emergency Procedures	☐ Yes ☐ No	☐ Yes ☐ No	
Vendor Contacts	☐ Yes ☐ No	☐ Yes ☐ No	
Warranty Information	☐ Yes ☐ No	☐ Yes ☐ No	

3. COMPLIANCE ASSESSMENT

Requirement	Compliant	Evidence	Comments
Electrical Codes	☐ Yes ☐ No ☐ Partial		
Safety Standards	☐ Yes ☐ No ☐ Partial		
Environmental Regulations	☐ Yes ☐ No ☐ Partial		
Insurance Requirements	☐ Yes ☐ No ☐ Partial		
Internal Standards	☐ Yes ☐ No ☐ Partial		
Manufacturer Recommendations	☐ Yes ☐ No ☐ Partial		
Industry Best Practices	☐ Yes ☐ No ☐ Partial		

4. SYSTEM CAPACITY AND LOADING

System	Design Capacity	Current Peak Load	Avg. Load	% Utilization	Growth Capacity	Comments
Main Utility Service						
Generator Capacity						
UPS Capacity						
Battery Runtime						

Main Distribution						
Critical Distribution						
Cooling Capacity						

5. RELIABILITY ASSESSMENT

System	Annual Availability	Outage Events	MTBF	MTTR	Reliability Trend	Comments
Utility Power					☐ Improved ☐ Steady ☐ Declined	
Generator System					☐ Improved ☐ Steady ☐ Declined	
UPS System					☐ Improved ☐ Steady ☐ Declined	
Transfer System					☐ Improved ☐ Steady ☐ Declined	
Overall Power System					☐ Improved ☐ Steady ☐ Declined	

6. MAINTENANCE PROGRAM EVALUATION

Aspect	Rating	Observations	Recommendations
Preventive Maintenance Schedule	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Maintenance Procedure Quality	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Maintenance Documentation	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Spare Parts Management	☐ Excellent ☐ Good ☐ Fair ☐ Poor		

Vendor Service Quality	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Staff Competency	☐ Excellent ☐ Good ☐ Fair ☐ Poor		
Test Equipment Adequacy	☐ Excellent ☐ Good ☐ Fair ☐ Poor		

Power System Maintenance SOP Completion

7. RISK ASSESSMENT (Continued)

Risk Category	Current Risk Level	Mitigation Effectiveness	Residual Risk	Action Required
Generator Failure	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
UPS Failure	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
Battery System Failure	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
Cooling System Failure	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
Human Error	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
Fire/Environmental	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	
Physical Security	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	☐ High ☐ Medium ☐ Low	

8. ENERGY EFFICIENCY ANALYSIS

Parameter	Current Value	Previous Year	Benchmark	Trend	Comments
PUE (Power Usage Effectiveness)				☐ Improved ☐ Steady ☐ Declined	
Generator Efficiency (%)				☐ Improved ☐ Steady ☐ Declined	
UPS Efficiency (%)				☐ Improved ☐ Steady ☐ Declined	
HVAC Efficiency (EER)				☐ Improved ☐ Steady ☐ Declined	
Energy Cost per kWh (\$)				☐ Improved ☐ Steady ☐ Declined	
Carbon Footprint (CO ₂ tons)				☐ Improved ☐ Steady ☐ Declined	
Renewable Energy Mix (%)				☐ Improved ☐ Steady ☐ Declined	

9. LIFECYCLE ANALYSIS AND REPLACEMENT PLANNING

Equipment Type	Installation Date	Expected Life	Remaining Life	Replacement Budget	Replacement Priority	Comments
UPS System					☐ High ☐ Medium ☐ Low	
Generator					☐ High ☐ Medium ☐ Low	
Battery System					☐ High ☐ Medium ☐ Low	
Transfer Switches					☐ High ☐ Medium ☐ Low	
Switchgear					☐ High ☐ Medium ☐ Low	
PDU's/RPP's					☐ High ☐ Medium ☐ Low	
Cooling Equipment					☐ High ☐ Medium ☐ Low	

10. TRAINING AND STAFFING ASSESSMENT

Aspect	Current Status	Adequacy	Recommendations
Operations Staff Headcount		☐ Adequate ☐ Inadequate	
Maintenance Staff Headcount		☐ Adequate ☐ Inadequate	
Technical Competency Levels		☐ Adequate ☐ Inadequate	
Training Program Coverage		☐ Adequate ☐ Inadequate	
Certifications		☐ Adequate ☐ Inadequate	
Emergency Response Training		☐ Adequate ☐ Inadequate	
Documented Procedures		☐ Adequate ☐ Inadequate	

11. STRATEGIC RECOMMENDATIONS

System Reliability Improvements

- 1.
- 2.
- 3.
- 4.
- 5.

Energy Efficiency Initiatives

- 1.
- 2.
- 3.
- 4.
- 5.

Maintenance Program Enhancements

- 1.
- 2.
- 3.
- 4.
- 5.

Technology Upgrades

- 1.
- 2.
- 3.
- 4.
- 5.

Risk Mitigation Actions

- 1.
- 2.
- 3.
- 4.
- 5.

12. BUDGET PROJECTIONS

Category	Current Year Actual (\$)	Next Year Projection (\$)	Justification
Preventive Maintenance			
Corrective Maintenance			
Spare Parts			
Equipment Replacement			
Training			
Consulting Services			
Energy Costs			
Efficiency Improvements			
Total			

13. EXECUTIVE SUMMARY

Key Findings

Performance Metrics

Critical Recommendations

Budget Summary

Strategic Outlook

14. APPROVALS

Position	Name	Signature	Date
Facility Manager			
Operations Director			
Maintenance Manager			
Safety Officer			
Finance Director			
Executive Approval			

15.3.2 Power Incident Report Form

BITCOIN MINING FACILITY POWER INCIDENT REPORT FORM

Facility Name: _____
Date of Incident: _____
Time of Incident: _____
Report Completed By: _____
Position: _____
Date Completed: _____

1. INCIDENT TYPE

- ☐ ☐ Complete Power Outage
- ☐ ☐ Partial Power Outage
- ☐ ☐ UPS Failure
- ☐ ☐ Generator Failure
- ☐ ☐ Transfer Switch Failure
- ☐ ☐ Battery System Issue
- ☐ ☐ Power Quality Event
- ☐ ☐ Equipment Damage
- ☐ ☐ Fire/Explosion
- ☐ ☐ Human Error
- ☐ ☐ Other: _____

2. INCIDENT DURATION

Start Time: _____
End Time: _____
Total Duration: _____

3. SYSTEMS AFFECTED

System	Impact Level	Description of Impact
☐ UPS	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Generator	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Transfer Switches	☐ None ☐ Minor ☐ Major ☐ Complete	

☐ Distribution System	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Cooling System	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Mining Equipment	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Monitoring Systems	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Safety Systems	☐ None ☐ Minor ☐ Major ☐ Complete	
☐ Other: _____	☐ None ☐ Minor ☐ Major ☐ Complete	

4. OPERATIONAL IMPACT

Impact Type	Details	Estimated Cost
☐ Mining Downtime		\$
☐ Equipment Damage		\$
☐ Repair Costs		\$
☐ Lost Revenue		\$
☐ Additional Labor		\$
☐ Other Costs		\$
Total Estimated Impact		\$

5. INCIDENT DESCRIPTION

Sequence of Events

Initial Observations

Immediate Actions Taken

Prior Warning Signs (if any)

6. ROOT CAUSE ANALYSIS

Primary Cause

Contributing Factors

Underlying Conditions

Root Cause Category

- ☐ ☐ Equipment Failure
- ☐ ☐ Maintenance Related
- ☐ ☐ Design/Installation Issue
- ☐ ☐ Procedural Issue
- ☐ ☐ Training Issue
- ☐ ☐ External Event
- ☐ ☐ Human Error
- ☐ ☐ Unknown
- ☐ ☐ Other: _____

7. RESPONSE EVALUATION

Response Aspect	Rating	Comments/Observations
Initial Detection	☐ Excellent ☐ Good ☐ Fair ☐ Poor	
Response Time	☐ Excellent ☐ Good ☐ Fair ☐ Poor	
Emergency Procedures	☐ Excellent ☐ Good ☐ Fair ☐ Poor	

Staff Performance	☐ Excellent ☐ Good ☐ Fair ☐ Poor	
Equipment Performance	☐ Excellent ☐ Good ☐ Fair ☐ Poor	
Communications	☐ Excellent ☐ Good ☐ Fair ☐ Poor	
Recovery Process	☐ Excellent ☐ Good ☐ Fair ☐ Poor	

8. CORRECTIVE ACTIONS

Action Item	Responsibility	Target Date	Status	Follow-up Required
		☐ Planned ☐ In Progress ☐ Complete	☐ Yes ☐ No	
		☐ Planned ☐ In Progress ☐ Complete	☐ Yes ☐ No	
		☐ Planned ☐ In Progress ☐ Complete	☐ Yes ☐ No	
		☐ Planned ☐ In Progress ☐ Complete	☐ Yes ☐ No	
		☐ Planned ☐ In Progress ☐ Complete	☐ Yes ☐ No	

9. PREVENTIVE MEASURES

Measure	Implementation Approach	Timeline	Resource Requirements	Priority
			☐ High ☐ Medium ☐ Low	
			☐ High ☐ Medium ☐ Low	
			☐ High ☐ Medium ☐ Low	
			☐ High ☐ Medium ☐ Low	
			☐ High ☐ Medium ☐ Low	

10. LESSONS LEARNED

11. SUPPORTING DOCUMENTATION

- ☐ ☐ Event Logs
- ☐ ☐ Alarm Records
- ☐ ☐ Maintenance Records
- ☐ ☐ Photos
- ☐ ☐ Video
- ☐ ☐ Witness Statements
- ☐ ☐ Vendor Reports
- ☐ ☐ Test Results
- ☐ ☐ Other: _____

12. REPORT DISTRIBUTION

Position	Name	Date Distributed
Facility Manager		
Operations Director		
Maintenance Manager		
Safety Officer		
IT/OT Manager		
Executive Team		
External Parties		

13. REPORT APPROVAL

Position	Name	Signature	Date
Incident Investigator			
Department Manager			
Operations Director			
Safety Officer			

15.3.3 Power Equipment Warranty Tracking Form

BITCOIN MINING FACILITY POWER EQUIPMENT WARRANTY TRACKING FORM

Facility Name: _____

Form Last Updated: _____

Maintained By: _____

1. UPS SYSTEM WARRANTY INFORMATION

Parameter	UPS System 1	UPS System 2	UPS System 3
Manufacturer			
Model			
Serial Number			
Installation Date			
Warranty Start Date			
Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Coverage Details			
Service Level	☐ Next Day ☐ Same Day ☐ 4hr	☐ Next Day ☐ Same Day ☐ 4hr	☐ Next Day ☐ Same Day ☐ 4hr
Vendor Contact			
Contract Number			
Renewal Date			
Renewal Cost			
Notes			

2. GENERATOR SYSTEM WARRANTY INFORMATION

Parameter	Generator 1	Generator 2	Generator 3
Manufacturer			
Model			
Serial Number			
Installation Date			
Warranty Start Date			
Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Coverage Details			
Hour Limitation			
Service Provider			
Vendor Contact			
Contract Number			
Renewal Date			
Renewal Cost			
Notes			

3. BATTERY SYSTEM WARRANTY INFORMATION

Parameter	Battery System 1	Battery System 2	Battery System 3
Manufacturer			
Model			
Installation Date			
Warranty Start Date			

Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Prorated Terms			
Coverage Details			
Vendor Contact			
Contract Number			
Renewal Options			
Disposal Requirements			
Notes			

4. TRANSFER SWITCH WARRANTY INFORMATION

Parameter	ATS 1	ATS 2	ATS 3
Manufacturer			
Model			
Serial Number			
Installation Date			
Warranty Start Date			
Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Coverage Details			
Service Provider			
Vendor Contact			
Contract Number			

Renewal Date			
Renewal Cost			
Notes			

5. DISTRIBUTION EQUIPMENT WARRANTY INFORMATION

Parameter	Main Switchgear	PDUs/RPPs	Branch Panels
Manufacturer			
Model/Type			
Serial Number			
Installation Date			
Warranty Start Date			
Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Coverage Details			
Service Provider			
Vendor Contact			
Contract Number			
Renewal Date			
Renewal Cost			
Notes			

6. COOLING SYSTEM WARRANTY INFORMATION

Parameter	CRAC/CRAH Units	Chillers	Air Handlers
Manufacturer			
Model			
Serial Number			
Installation Date			
Warranty Start Date			
Warranty End Date			
Warranty Type	☐ Standard ☐ Extended	☐ Standard ☐ Extended	☐ Standard ☐ Extended
Coverage Details			
Service Provider			
Vendor Contact			
Contract Number			
Renewal Date			
Renewal Cost			
Notes			

7. WARRANTY CLAIM HISTORY

Equipment	Claim Date	Claim Number	Issue Description	Resolution	Cost Covered	Out of Pocket Cost	Status
				\$	\$	☐ Open ☐ Closed	
				\$	\$	☐ Open ☐ Closed	

				\$	\$	☐ Open ☐ Closed	
				\$	\$	☐ Open ☐ Closed	
				\$	\$	☐ Open ☐ Closed	
				\$	\$	☐ Open ☐ Closed	

8. WARRANTY EXPIRATION CALENDAR

Equipment	Type	Serial Number	Warranty End Date	Days Remaining	Renewal Required	Priority
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	

				☐ Yes ☐ No	☐ High ☐ Medium ☐ Low	
--	--	--	--	--	--	--

9. ASSOCIATED DOCUMENT LOCATIONS

Document Type	Storage Location	Access Method	Responsible Party
Original Warranty Documents			
Service Contracts			
Purchase Orders			
Inspection Records			
Maintenance Logs			
Claim Documentation			
Vendor Communications			

Last Updated By: _____

Date: _____

Reviewed By: _____

Date: _____